SMART WALKING STICK FOR VISUALLY IMPAIRED NAVIGATION WITH AI INTEGRATION

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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology specialized in Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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..... 30/Aug/2025.....

Signature of the Supervisor Date

(Dr. Junius Anjana)

ABSTRACT

Visual impairment affects approximately 2.2 billion people worldwide, creating significant challenges in independent navigation and environmental awareness. Traditional assistive devices like white canes provide limited information about complex environmental landmarks such as crosswalks, building entrances, and traffic signals, which are crucial for safe navigation. This component of the Smart Walking Stick project addresses the critical need for intelligent landmark recognition to enhance mobility independence for visually impaired individuals.

The proposed landmark recognition system integrates ESP32-CAM modules with TensorFlow Lite machine learning models to provide real-time identification of environmental landmarks. The system captures visual data through the ESP32-CAM, processes it using a trained TinyML model for landmark classification, and provides immediate feedback through vibration patterns and audio alerts. The component focuses on recognizing essential navigation landmarks including crosswalks, traffic lights, building entrances, and pedestrian signals.

The research employs computer vision techniques, edge computing, and human-computer interaction principles to develop a lightweight, power-efficient solution that operates in real-time. The system ensures minimal latency while maintaining high accuracy in landmark detection, making it suitable for integration into a portable assistive device. Expected outcomes include improved navigation safety, enhanced environmental awareness, and increased independence for visually impaired users in urban environments.

Keywords: Landmark Recognition, Computer Vision, TinyML, ESP32-CAM, Assistive Technology, Visual Impairment

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ETA	Electronic Travel Aid
FPS	Frames Per Second
IoT	Internet of Things
MTBF	Mean Time Between Failures
PCB	Printed Circuit Board
Tiny ML	Tiny Machine Learning
FR	Functional Requirements
NFR	Non-Functional Requirement

1. INTRODUCTION

Visual impairment is one of the most challenging disabilities that affects personal mobility and independent living. The World Health Organization reports that nearly 2.2 billion people around the world experience some form of vision impairment, with at least 1 billion cases being preventable or still unaddressed [1]. For those living with visual impairment, moving through busy and complex urban spaces involves numerous difficulties, including avoiding obstacles, identifying landmarks, and maintaining spatial awareness.

Conventional aids such as white canes and guide dogs provide valuable support but remain limited in scope. A white cane mainly detects obstacles on the ground through direct contact and offers little awareness of overhead dangers, the layout of the surroundings, or the location of important landmarks [2]. Guide dogs, while effective in enhancing mobility, require extensive training and represent a considerable financial commitment for both acquisition and upkeep.

Recent developments in Internet of Things (IoT), computer vision, and machine learning have paved the way for smarter assistive technologies. Advances in edge computing and TinyML (Tiny Machine Learning) now allow sophisticated AI models to run on small, low-power devices, enabling real-time analysis without depending on continuous internet access [3].

Among these advancements, landmark recognition has become a key focus for improving navigation support. Features such as pedestrian crossings, traffic lights, doorways, and building entrances serve as essential reference points, helping visually impaired individuals orient themselves and make safe, informed decisions while traveling [4]. Most existing assistive technologies emphasize obstacle detection but fall short in delivering information about essential environmental landmarks. By combining computer vision with embedded hardware, it becomes possible to close this gap through systems capable of recognizing, classifying, and conveying landmark details in real time. The ESP32-CAM microcontroller, equipped with an integrated

camera and Wi-Fi connectivity, offers a suitable platform for such applications. Its low energy requirements, small form factor, and adequate processing power make it well-suited for running lightweight machine learning models [5].

The focus of this research is the design of a landmark recognition system built on ESP32-CAM modules and TensorFlow Lite models, aimed at detecting environmental landmarks and delivering immediate feedback to individuals with visual impairments. The system represents significant advancement in assistive technology by combining edge computing, computer vision, and human-computer interaction principles to create a practical, affordable, and effective navigation aid.

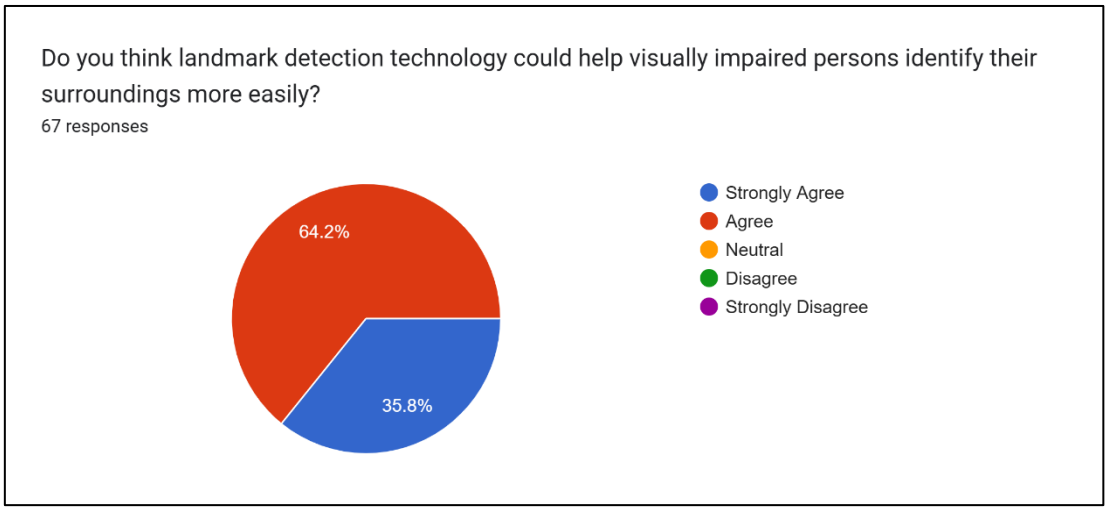


Figure 1.1 How people think landmark detection for visual impaired persons

1.1 Background & Literature Survey

1.1.1 Assistive technologies for visual impairment

Over the past several decades, assistive technologies for visually impaired individuals have undergone substantial evolution. Initial approaches largely relied on mechanical aids, including white canes and guide dogs. Introduced in the 1930s, the white cane continues to be the most used mobility tool, valued for its simplicity, cost-effectiveness, and ability to detect obstacles at ground level [6]. Despite these

advantages, it is limited in conveying information about overhead hazards and broader environmental features.

Electronic travel aids (ETAs) emerged in the 1960s with devices like the Sonic Torch and the Path sounder, which used ultrasonic waves to detect obstacles [7]. These early systems demonstrated the potential of technology to enhance mobility but were limited by their single-function nature and high cost. The advancement of microelectronics and sensor technology has led to more sophisticated systems that integrate multiple sensing modalities.

1.1.2 Computer vision in assistive technology

Computer vision has become a vital component in the advancement of assistive technologies. Systems for object recognition have been effectively applied in diverse areas, including text reading and identification of everyday items [8]. Applying computer vision to landmark recognition, however, introduces specific challenges, such as variations in lighting, weather conditions, and architectural designs.

Deep learning approaches, especially Convolutional Neural Networks (CNNs), have shown outstanding success in solving complex computer vision problems such as image classification and object detection [9]. Their ability to automatically learn hierarchical feature representations from raw image data has significantly improved the accuracy and reliability of recognition systems.

In recent years, the introduction of lightweight architecture such as MobileNet and EfficientNet has further advanced this field. These models are specifically optimized for efficiency, enabling deployment on devices with limited processing power and memory while maintaining competitive levels of accuracy [10]. Such architectures make it feasible to bring deep learning out of high-performance computing environments and into embedded systems, opening new opportunities for real-time assistive applications.

1.1.3 Edge computing and TinyML

Edge computing has transformed the way machine learning models are deployed on embedded systems. TinyML, a rapidly expanding field dedicated to running machine learning on microcontrollers and other resource-limited devices, has opened new opportunities for real-time AI applications [11].

TensorFlow Lite, Google's platform for executing machine learning models on mobile and embedded hardware, has become a widely adopted tool in edge AI. Its optimization strategies, such as model quantization and pruning, enable the efficient operation of complex models on devices with constrained memory and processing power [12].

1.1.4 ESP32-CAM in computer vision applications

The ESP32-CAM development board has become widely used in computer vision projects due to its built-in camera, Wi-Fi connectivity, and adequate processing capabilities for lightweight machine learning models [13]. Its affordability and compact design make it particularly well-suited for prototyping and implementing embedded vision systems.

Studies have shown that the ESP32-CAM can successfully perform a range of computer vision tasks, including face recognition, object detection, and image classification [14]. Its capability to capture images, process data locally, and transmit results wirelessly makes it an effective platform for real-time assistive applications.

1.1.5 Landmark recognition systems

Existing landmark recognition systems have primarily focused on large-scale applications such as autonomous vehicles and augmented reality. These systems often rely on extensive computational resources and high-resolution cameras, making them unsuitable for portable assistive devices [15].

Some research has been conducted on landmark recognition for navigation assistance, but most solutions require smartphone-based processing or cloud connectivity, which may not be practical for real-time applications or users with limited access to modern smartphones [16].

1.1.6 Human computer interaction in assistive technology

The design of effective human-computer interfaces for visually impaired users requires careful consideration of alternative sensory modalities. Haptic feedback, audio cues, and vibrotactile patterns have been extensively studied as means of conveying information to visually impaired users [17].

Research has shown that vibrotactile feedback can effectively communicate spatial and categorical information when properly designed [18]. The development of intuitive feedback patterns that can convey landmark information without overwhelming the user represents a significant design challenge in assistive technology.

1.2 Research gap

Despite significant advances in assistive technology and computer vision, several critical gaps remain in current landmark recognition systems for visually impaired individuals:

1.2.1 Limited real-time processing capabilities

Most existing landmark recognition systems rely on smartphone-based processing or cloud connectivity, introducing latency that can be problematic for real-time navigation assistance. The need for constant internet connectivity also limits the usability of these systems in areas with poor network coverage. Current solutions often prioritize accuracy over speed, making them unsuitable for dynamic navigation scenarios where immediate feedback is crucial.

1.2.2 Lack of embedded, portable solutions

Existing assistive technologies often require separate devices such as smartphones or tablets, increasing complexity and cost for users. The integration of landmark recognition capabilities directly into mobility aids has been limited due to the computational requirements of traditional computer vision algorithms. This gap represents a significant barrier to widespread adoption of intelligent assistive technologies.

1.2.3 Insufficient focus on environmental landmarks

While many assistive technologies focus on obstacle detection, there is limited research on recognizing specific environmental landmarks that are crucial for navigation orientation. Current systems often treat all detected objects equally, without prioritizing landmarks that provide spatial context and navigation reference points.

1.2.4 Limited customization and adaptability

Current landmark recognition systems often struggle to adapt to varying environments and individual user needs. The models are generally trained on standard datasets, which may not accurately represent the specific landmarks or architectural characteristics found in different geographic areas.

1.2.5 Inadequate feedback mechanisms

Many existing systems depend primarily on audio feedback, which can be ineffective in noisy settings or for users with hearing difficulties. The design and implementation of multimodal feedback systems, integrating haptic, auditory, and other sensory channels, remain largely unexplored.

1.2.6 Cost and accessibility barriers

Many existing solutions are expensive and require technical expertise to operate, limiting their accessibility to users who could benefit from them. The lack of

affordable, easy-to-use landmark recognition systems represents a significant barrier to improving mobility independence for visually impaired individuals.

In summary, these gaps highlight the need for a lightweight, embedded landmark recognition system that operates in real-time, focuses on environmental landmarks, offers adaptive functionality, provides multimodal feedback, and remains accessible and affordable. Table 1.1 below provides a comparison of existing smart stick systems with the proposed system, highlighting how current solutions address or fail to address these critical gaps.

Table 1.1 comparison of existing smart stick systems with the proposed system

Feature / System	We WALK Smart Cane (Research 1)	Smart Cane IIT Delhi (Research 2)	IoT-Based Smart Stick (Research 3)	A IoT Smart Stick (Research 4)	Proposed Research
Real-time Landmark Recognition	✕	✕	✕	✕	✓
Embedded Processing	✕	✓	✓	✕	✓
Environmental Landmark Detection	✕	✕	✕	✕	✓
Multimodal Feedback (audio + haptic)	✓ (audio only)	✓ (vibration only)	✓ (audio only)	✓ (audio only)	✓ (audio + haptic)
Portable / Integrated	✓	✓	✓	✓	✓
Affordable / Accessible	✕	✓	✓	✕	✓

1.3 Research problem

Individuals with visual impairments encounter considerable difficulties in recognizing and understanding environmental landmarks that are vital for safe and independent navigation. Conventional mobility aids, such as white canes and guide dogs, are effective for detecting obstacles at a basic level but do not convey information about important environmental features, including crosswalks, traffic signals, building entrances, and pedestrian crossings. The lack of landmark recognition functionality in existing assistive technologies gives rise to several key challenges:

1.3.1 Navigation uncertainty

Without the ability to identify landmarks, visually impaired individuals often experience uncertainty about their precise location and orientation within their environment. This uncertainty can lead to difficulty in maintaining directional awareness, challenges in locating specific destinations, increased reliance on assistance from others, and reduced confidence in independent navigation.

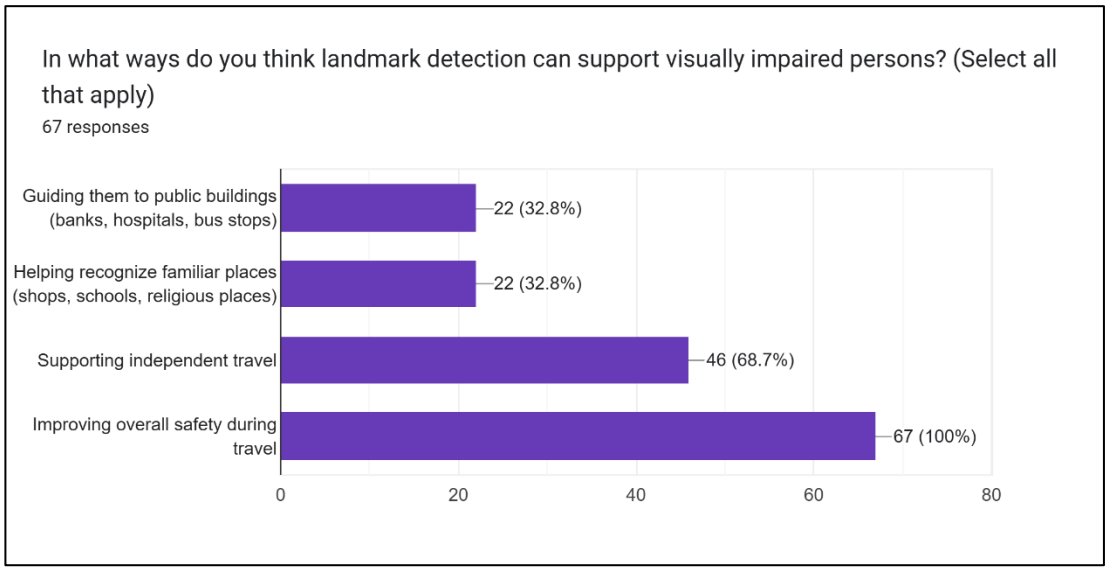


Figure 1.2 How supports visually impaired users in safe and efficient navigation.

1.3.2 Safety concerns

The inability to recognize important safety landmarks such as crosswalks and traffic signals poses significant safety risks including increased risk of accidents when

crossing streets, difficulty in identifying safe crossing points, inability to detect changes in traffic patterns or road conditions, and limited awareness of potential hazards in the immediate environment.

1.3.3 Limited environmental awareness

Current assistive technologies provide minimal information about the broader environmental context, resulting in lack of spatial reference points for orientation, inability to identify important buildings or facilities, limited understanding of environmental layout and structure, and reduced situational awareness in complex environments.

1.3.4 Technology limitations

Existing landmark recognition solutions face several technological limitations including dependency on internet connectivity for cloud-based processing, high latency in providing feedback, requirement for expensive hardware platforms, limited battery life and portability issues, and inadequate performance in varying lighting conditions.

1.3.5 User experience challenges

Current systems often fail to provide intuitive and accessible user interfaces, with complex setup and operation procedures, overwhelming or unclear feedback mechanisms, limited customization options for different user needs, and inadequate training and support resources. The research problem can be formulated as follows:

"How can we develop an intelligent, embedded landmark recognition system that provides real-time identification of environmental landmarks to enhance navigation safety and independence for visually impaired individuals, while addressing the limitations of current assistive technologies in terms of cost, portability, and user accessibility?"

2. OBJECTIVES

2.1 Main objective

To develop an intelligent landmark recognition system using ESP32-CAM and TensorFlow Lite machine learning models that provide real-time identification and feedback of environmental landmarks to enhance navigation safety and independence for visually impaired individuals.

2.2 Specific objectives

2.2.1 Machine learning model development

The first objective is to develop and optimize a machine learning model capable of accurate landmark classification. A TensorFlow Lite model will be trained and tuned to achieve a minimum of 85 percent accuracy on test datasets. Model quantization techniques will be applied to reduce the overall model size while maintaining acceptable performance on embedded hardware. A dedicated dataset will be created, consisting of environmental landmarks such as crosswalks, traffic signals, building entrances, and pedestrian crossings. Model validation will be performed under diverse lighting conditions and environmental scenarios to ensure robustness.

2.2.2 ESP32-CAM implementation

Another goal is to configure the ESP32-CAM for real-time image acquisition and preprocessing, allowing landmarks to be detected directly on the device. TinyML inference will be incorporated into the ESP32-CAM to enable efficient and lightweight classification. Camera settings will be optimized for adaptability in varying lighting conditions, while efficient image processing pipelines will be developed to minimize latency and support smooth real-time performance.

2.2.3 Communication system integration

A reliable communication framework must be established to connect the ESP32-CAM with the Arduino Uno controller. This objective includes implementing a serial communication protocol to support consistent data transfer of landmark detection results. Mechanisms will be designed to ensure reliable and low-latency communication between system components. In addition, error handling and recovery methods will be integrated to maintain communication stability even in cases of system failure or interference.

2.2.4 Feedback system development

The feedback system will be designed to deliver intuitive cues to users, allowing them to respond quickly to detected landmarks. Vibrotactile feedback patterns will be created for different landmark categories, while audio alerts will be developed with customizable sounds to suit user preferences. An intuitive mapping between detected landmarks and feedback types will be implemented to ensure users can easily distinguish between signals. To assess usability, the feedback system will be tested with target user groups, including visually impaired participants.

2.2.5 System integration and testing

Once the individual components are developed, they will be integrated into the overall smart walking stick system. Extensive testing will be conducted in real-world settings to assess the system's performance under practical conditions. Evaluation will focus on key metrics, including accuracy, response time, and reliability. User acceptance testing will also be conducted with visually impaired individuals to determine usability and overall effectiveness in assisting navigation.

2.2.6 Performance optimization

The final objective is to optimize system performance for long-term and real-world usage. Power consumption will be minimized to extend battery life during outdoor activities. Processing latency will be reduced to ensure near real-time feedback, while system robustness will be improved to withstand different environmental conditions. Additionally, adaptive algorithms will be implemented to enhance performance over time, allowing the system to learn and adjust to user needs and changing surroundings.

3. METHODOLOGY

3.1 Overall system description

The landmark recognition component represents a central feature of the Smart Walking Stick system, providing real-time detection and identification of key environmental landmarks. By recognizing elements such as crosswalks, traffic signals, and building entrances, the system supports visually impaired individuals in navigating their surroundings with greater safety and independence.

This functionality is achieved through the integration of computer vision algorithms, Figure 3.1 presents the system architecture, highlighting the interaction between hardware components, recognition models, and feedback modules.

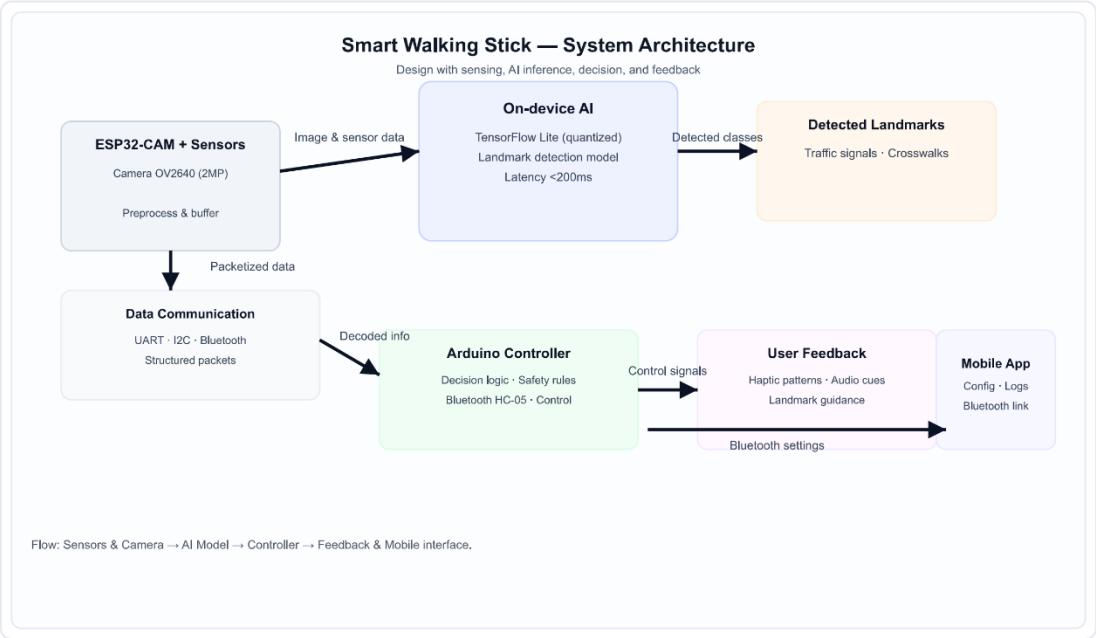


Figure 3.1 System Architectural Diagram

The processed information is then communicated to the user through feedback mechanisms, making the device both portable and user-friendly. Together, these elements form a compact yet powerful assistive technology solution, designed to operate reliably under real-world conditions while remaining accessible in terms of cost and usability.

3.2 Development methodology

The project follows an iterative development approach based on the Rapid Application Development (RAD) methodology, allowing for quick prototyping and continuous refinement based on user feedback and testing results.

3.2.1 Ghantt Chart

The Gantt chart for the "Smart Walking Stick for Visually Impaired" project outlines a structured timeline from August 2025 to June 2026. It details key phases, including planning, design, and model development, with specific tasks such as project initiation, literature review, hardware/software architecture design, and mobile app development. Each task is color-coded and scheduled to ensure progress, quick prototyping, and continuous refinement based on user feedback and testing results.

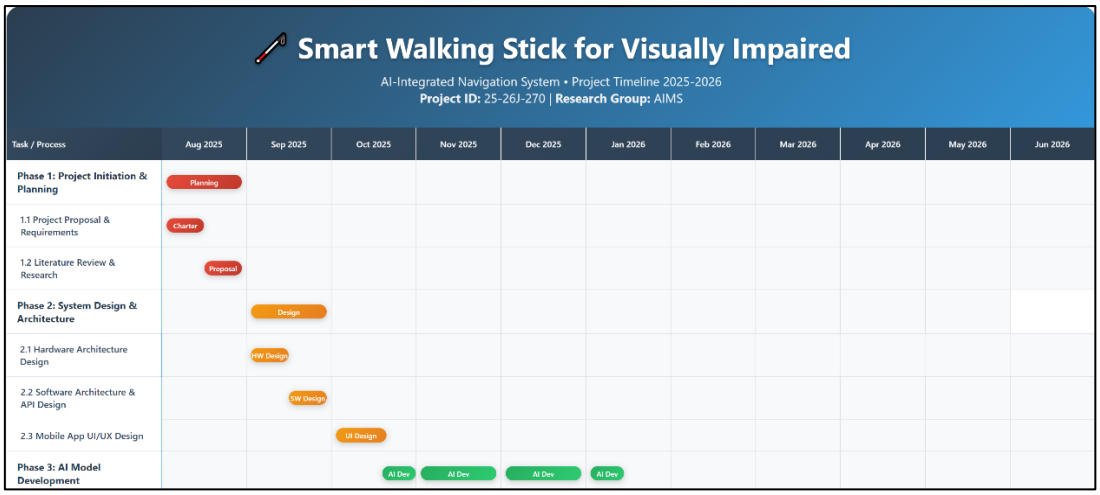


Figure 3.2 Gantt Chart for the entire research project

Full hosted Ghantt chart for the whole year - https://ghant-chart-25-26j-270.netlify.app/

3.2.2 Work Breakdown Structure

Phase 1: Research and Planning (Weeks 1-4)

- Literature review and technology assessment
- Dataset collection and preparation
- Hardware component selection and procurement
- Development environment setup

Phase 2: Model Development (Weeks 5-12)

- Dataset creation and annotation
- Model architecture design and training
- Model optimization and quantization
- Performance evaluation and validation

Phase 3: Hardware Implementation (Weeks 13-20)

- ESP32-CAM programming and configuration
- Communication protocol implementation
- Hardware integration and testing
- Power optimization and reliability testing

Phase 4: System Integration (Weeks 21-28)

- Component integration with main system
- Feedback system development and testing
- User interface development
- Comprehensive system testing

Phase 5: Evaluation and Refinement (Weeks 29-36)

- Real-world testing and validation
- User acceptance testing and deployment
- Performance optimization and documentation

3.3 Technical implementation approach

3.3.1 Machine learning model development

The development of the machine learning model begins with the creation of a comprehensive and diverse dataset. Images are systematically collected from a wide range of environmental landmarks, such as crosswalks, traffic lights, building entrances, and pedestrian crossing signals. To ensure that the model can perform reliably in real-world scenarios, the dataset captures variations in lighting conditions, weather patterns, and architectural styles, reflecting the diversity of urban environments.

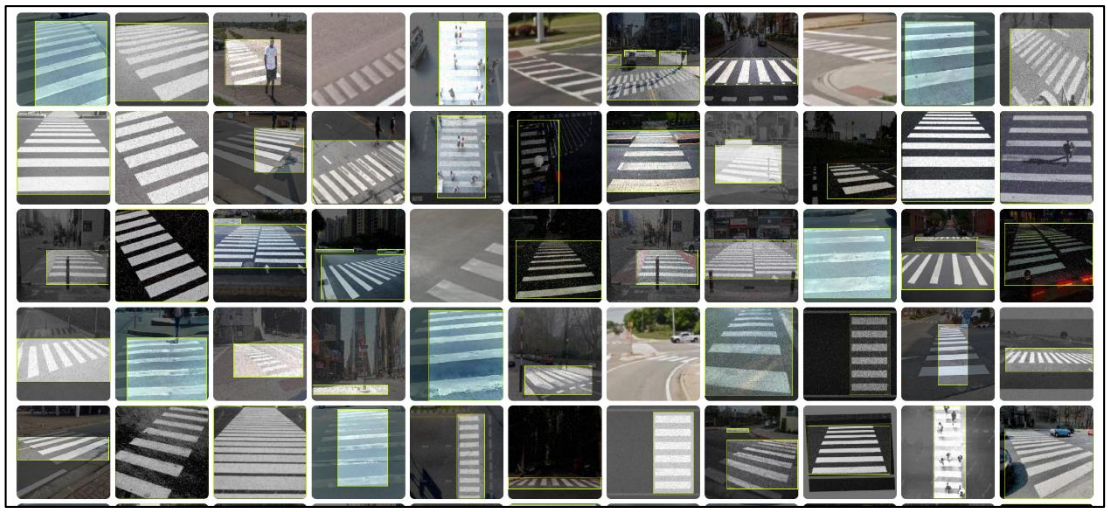


Figure 3.3 Sample data set image of crosswalks

Data augmentation techniques such as rotation, scaling, brightness adjustment, and flipping are applied to artificially increase the dataset size and introduce further variability. This helps the model generalize better to previously unseen conditions. Careful attention is also given to balancing the dataset, ensuring that each landmark category has sufficient representation to prevent bias and improve classification accuracy. The resulting dataset provides a strong foundation for training a robust and reliable landmark recognition model.

For model architecture, MobileNetV2 is chosen as the foundation because of its lightweight design and efficiency, making it well-suited for performing real-time inference on embedded devices with limited computational resources. Its streamlined structure allows for a balance between high accuracy and low latency, which is essential for on-device landmark recognition. Transfer learning with pre-trained ImageNet weights accelerates model training, and custom classification layers are designed to support landmark-specific recognition. To optimize performance on resource-constrained hardware, quantization techniques are applied, reducing both the model size and inference time.

The training process begins with stratified splitting of the dataset to ensure that the proportions of each landmark category are maintained in both the training and validation sets. Cross-validation techniques are applied to evaluate the model's robustness across multiple data subsets, providing a more reliable estimate of its performance.

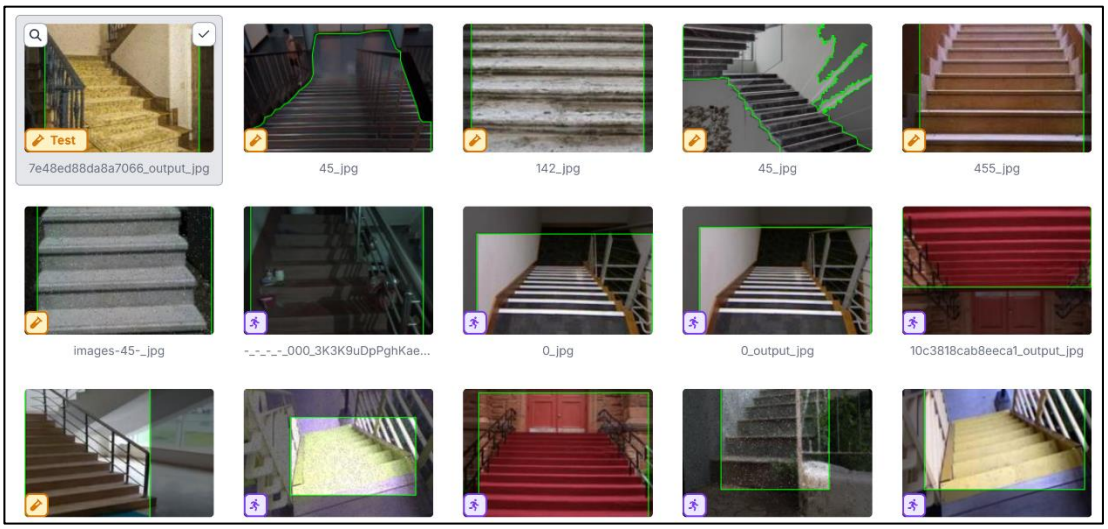


Figure 3.4 Staire dataset sample image

Additionally, regularization methods, such as dropout and weight decay, are incorporated during training to reduce the risk of overfitting and improve the model's generalization to unseen data. Hyperparameter tuning is conducted through a combination of grid search and random search approaches to achieve optimal model performance.

3.3.2 ESP32-CAM implementation

Hardware Configuration: The ESP32-CAM is configured to deliver reliable image capture in varying conditions. Camera settings are adjusted for optimal image quality, with automatic exposure and white balance control to adapt to changing environments. Memory management techniques are implemented to handle image processing efficiently, while power management settings are optimized to extend battery life.

Software Development: On the software side, modules are developed for image capture and preprocessing, followed by integration of the TensorFlow Lite inference engine to support on-device recognition. Communication protocols are implemented for transmitting landmark information to the Arduino, while error handling and recovery mechanisms are added to maintain stable operation.

3.3.3 Communication system

The communication framework facilitates dependable data exchange between the ESP32-CAM and the Arduino. Serial communication is carefully configured and fine-tuned to ensure stable and consistent performance, allowing real-time transmission of image data and classification results for further processing or feedback. A structured data packet format is designed for transmitting landmark information clearly and efficiently. To safeguard against errors, acknowledgment mechanisms are included, along with timeout and retry methods to strengthen reliability during communication.

3.3.4 Feedback system design

Audio Alerts: The feedback system delivers information to the user through audio signals. Different landmark types are indicated by unique tone frequencies, while voice announcements provide more detailed descriptions. The audio output automatically adjusts its volume based on the surrounding noise level, ensuring clarity in varied environments. Furthermore, the system supports multiple languages to meet the needs of a wider user base.

3.4 Testing and validation methodology

Laboratory testing: Initial testing is conducted in a controlled environment with known landmarks. The system is benchmarked under different lighting conditions, subjected to stress tests for reliability, and evaluated for power consumption to optimize performance.

Field Testing: Trials are conducted in urban environments to evaluate system performance under real-world conditions. Testing covers a range of weather scenarios and lighting conditions to assess robustness. Visually impaired participants are involved in user acceptance testing to provide direct feedback on usability and effectiveness. The system's performance is also benchmarked against existing assistive technologies to highlight its advantages and potential improvements.

Evaluation Metrics: The system is evaluated using multiple quantitative and qualitative measures. Classification accuracy and precision are used to assess the reliability of landmark recognition. Real-time performance is measured through system response time and latency. Additionally, power consumption and battery life are monitored to ensure the system is practical for everyday use. User satisfaction and usability scores are collected to gauge the effectiveness of the system in supporting visually impaired individuals and identify areas for refinement.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

4.1.1 Core functionality

The core functionality of the system focuses on accurate image capture, landmark recognition, and real-time feedback delivery. The ESP32-CAM is required to capture high-resolution images at a minimum rate of two frames per second. The machine learning model must classify environmental landmarks with at least 85 percent accuracy, recognizing crosswalks, traffic signals, building entrances, and pedestrian crossings. Once a landmark is detected, the system must provide real-time feedback to the user within 500 milliseconds. In addition, all detection results must be reliably transmitted to the Arduino controller via serial communication for further processing and integration with other subsystems.

4.1.2 User interface requirements

The user interface emphasizes accessible and customizable feedback. Distinct vibrotactile patterns will be used to differentiate between landmark categories, while audio alerts will be generated with customizable volume settings. The system will support multiple feedback modes, including vibration only, audio only, and a combined option, giving users the flexibility to select their preferred method. Furthermore, users will have the option to enable or disable specific landmark categories according to personal preference. A mobile application will also provide battery status indicators, ensuring that users always remain informed about system power levels.

4.1.3 Communication requirements

Communication between components is designed to be both reliable and efficient. The system will maintain continuous and error-free serial communication with the Arduino controller while employing error detection and correction mechanisms to ensure accuracy. Data transmission rates will be supported at a minimum of 9600 baud, guaranteeing consistent performance. Beyond the wired interface, the system

will also integrate with the mobile application through Bluetooth connectivity, allowing extended functionality and user control.

4.1.4 Configuration requirement

System configuration allows users to personalize their experience. Detection sensitivity levels can be adjusted to meet different environmental conditions. Audio feedback will support multiple language options, enhancing accessibility for users from diverse backgrounds. Feedback patterns, whether vibrotactile or auditory, will also be customizable. Finally, all user preferences will be stored in non-volatile memory so that settings are retained even when the system is powered off.

4.2 Non-functional requirements

4.2.1 Performance requirements

The system must achieve a response time of no more than 500 milliseconds from image capture to feedback delivery. Landmark classification accuracy should remain above 85 percent, with the ESP32-CAM sustaining at least two frames per second during operation. The device should run continuously for a minimum of four hours on a single battery charge, while memory usage must not exceed 80 percent of the available capacity to ensure stability.

4.2.2 Reliability requirements

High reliability is essential for practical deployment. The system must achieve an availability rate of 99.5 percent during normal operation and a mean time between failures greater than 1000 hours. In the event of communication errors, recovery must occur within two seconds. Data transmission reliability is reinforced by ensuring the error rate does not exceed 0.1 percent. The system should also maintain dependable performance in a temperature range between 5°C and 40°C.

4.2.3 Usability requirements

The design places emphasis on ease of use. New users should be able to learn system operation within 30 minutes, and the feedback should remain intuitive enough to require no prior technical training. Setup and configuration must be completed in under

10 minutes, even by individuals with limited technical expertise. All user interface elements will adhere to accessibility standards to support inclusivity.

4.2.4 Portability and integration requirements

The system must remain lightweight and seamlessly integrated into the walking stick. The ESP32-CAM module should not exceed 50 grams, while the design must blend effectively with the overall hardware structure. Power consumption is limited to 500mA during normal operation to preserve battery efficiency. The system will also feature IPX4 water resistance for protection against splashing. Bluetooth communication is required to function reliably within a minimum range of 10 meters.

4.3 System requirements

4.3.1 Hardware requirements

The hardware platform consists of the ESP32-CAM development board with at least 4MB flash memory, paired with the OV2640 camera sensor supporting a resolution of 2 megapixels. An Arduino Uno R3 or compatible microcontroller is used for processing and integration, supported by the HC-05 Bluetooth module for wireless communication. User feedback is provided through vibration motors operating at a minimum of 3V and a piezo buzzer for audio signals. The system will be powered by a rechargeable lithium-ion battery with a minimum capacity of 2000mAh, supported by voltage regulation circuits for a stable power supply.

4.3.2 Software requirements

The hardware platform consists of the ESP32-CAM development board with at least 4MB flash memory, paired with the OV2640 camera sensor supporting a resolution of 2 megapixels. An Arduino Uno R3 or compatible microcontroller is used for processing and integration, supported by the HC-05 Bluetooth module for wireless communication. User feedback is provided through vibration motors operating at a minimum of 3V and a piezo buzzer for audio signals. The system will be powered by a rechargeable lithium-ion battery with a minimum capacity of 2000mAh, supported by voltage regulation circuits for a stable power supply.

5. BUDGET AND BUDGET JUSTIFICATION

5.1 Hardware costs

Table 5.1 Budget allocate for the landmark detection component

Component	Qty	Unit Price (LKR)	Total Cost (LKR)	Justification
ESP32-CAM Development Board	2	2,100	4,200	Multiple units for testing, development, and backup
Arduino Uno R3	2	2,550	5,100	Primary and backup controllers
HC-05 Bluetooth Module	2	350	700	Communication with mobile device
Vibration Motors (3V)	5	200	1000	Haptic feedback generation
Piezo Buzzer	3	50	150	Audio alert generation
Lithium-ion Battery (18650)	1	2,700	2,700	Power supply for system
Battery Holder and Charging Circuit	2	100	200	Safe battery management
Jumper Wires and Connectors	1 Set	350	350	System connectivity
Breadboards and PCB	3	100	300	Prototyping and final assembly
Enclosure and Mounting Hardware	1 Set	500	500	Weather Protection and integration
Hardware Subtotal			**15,200**	

5.2 Software and development costs

Table 5.2 Software and development costs

Item	Cost (LKR)	Justification
Arduino IDE and Libraries	Free	Open-source development environment
TensorFlow Lite	Free	Open-source machine learning framework
Image Dataset Creation	1,500	Data collection, annotation, and processing
Cloud Storage for Dataset	2,400	Google Drive Pro subscription for 6 months
Mobile App Development Tools	Free	Flutter framework is open source
Software Subtotal	**3,900**	

6. COMMERCIALIZATION

6.1 Target Market

The primary target market for the smart walking stick consists of visually impaired individuals, a population estimated at 2.2 billion worldwide, with at least 39 million experiencing complete blindness. Beyond individual users, the system is also intended for rehabilitation centers, hospitals, and disability support organizations, where it can be integrated into training and mobility assistance programs.

6.2 Value Proposition

The proposed solution offers a unique value proposition by being significantly more affordable than guide dogs and high-end electronic aids while still providing advanced functionality. Real-time landmark detection enhances user safety and independence, giving individuals greater confidence when navigating their surroundings. In addition, the system is lightweight, portable, and designed for ease of use, requiring no technical expertise from the user.

6.3 Competitive Advantage

The system differentiates itself from existing solutions through its ability to recognize landmarks rather than merely detecting obstacles. This capability adds meaningful context to navigation, making the device more useful in real-world situations. Furthermore, multimodal feedback through both haptic and audio cues improves usability and ensures accessibility across a wider range of users. By relying on low-cost hardware components such as the ESP32-CAM and Arduino, the device remains affordable without compromising performance.

6.5 Business Model

The business model focuses on direct sales of the device through established assistive technology vendors. Partnerships with disability councils, non-governmental organizations, and government programs will expand distribution and help reach communities in need. Such collaborations will also support subsidized access to ensure that cost is not a barrier to adoption.

6.6 Pricing Strategy

The pricing strategy is designed to maintain affordability, with an estimated cost of approximately LKR 10,000–15,000 per unit. To further improve accessibility, the project envisions subsidies or discounts through healthcare programs and disability support schemes. This approach ensures that the product remains within reach of the intended users while supporting long-term market penetration.

6.7 Future Expansion

Future development will focus on integrating GPS navigation to provide outdoor guidance and route planning. Additional features will be introduced through a mobile application, allowing customization of feedback modes and offering remote support services. With scaling of production, the project also aims to expand into international markets, contributing to global accessibility solutions for the visually impaired.

REFERENCES

[1] World Health Organization, "Blindness and vision impairment," WHO Fact Sheet, Geneva, Switzerland, updated Aug. 10, 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment. [Accessed: Aug. 31, 2025].

[2] S. Shoval, I. Ulrich, and J. Borenstein, "NavBelt and the Guide-Cane [obstacle-avoidance systems for the blind and visually impaired]," *IEEE Robotics & Automation Magazine*, vol. 10, no. 1, pp. 9-20, Mar. 2003, doi: 10.1109/MRA.2003.1191706.

[3] V. J. Reddi, C. Plancher, S. Kennedy, L. Leong, R. Banbury, D. Dreyer, M. Janapa-Reddi, A. Joshi, U. Ugalde, P. Whatmough, and M. Mattina, "Widening access to applied machine learning with TinyML," Harvard John A. Paulson School of Engineering and Applied Sciences, Technical Report, Cambridge, MA, USA, 2021.

[4] M. Bousbia-Salah, A. Redjati, M. Fezari, and M. Bettayeb, "An ultrasonic navigation system for blind people," in *Proc. IEEE International Conference on Signal Processing and Communications (ICSPC 2007)*, Dubai, United Arab Emirates, Nov. 24-27, 2007, pp. 1003-1006, doi: 10.1109/ICSPC.2007.4728524.

[5] Espressif Systems, "ESP32-CAM Development Board User Guide," ver. 1.2, Shanghai, China, 2020. [Online]. Available: https://docs.espressif.com/projects/esp-idf/en/latest/esp32/hw-reference/esp32/get-started-devkitc.html. [Accessed: Aug. 31, 2025].

[6] R. G. Golledge, "Geography and the disabled: a survey with special reference to vision impaired and blind populations," *Transactions of the Institute of British Geographers*, vol. 18, no. 1, pp. 63-85, 1993, doi: 10.2307/622307.

[7] C. Kärcher, L. Fenzlaff, D. Hartmann, S. Nagel, and E. König, "Sensory augmentation for the blind," *Frontiers in Human Neuroscience*, vol. 6, article 37, pp. 1-15, Feb. 2012, doi: 10.3389/fnhum.2012.00037.

[8] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. 3rd International Conference on Learning Representations (ICLR 2015)*, San Diego, CA, USA, May 7-9, 2015, pp. 1-14. [Online]. Available: https://arxiv.org/abs/1409.1556

[9] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems 25 (NIPS 2012)*, Lake Tahoe, NV, USA, Dec. 3-6, 2012, pp. 1097-1105.

[10] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNets: Efficient convolutional neural networks for mobile vision applications," in *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, Jun. 18-22, 2018, pp. 4510-4520, doi: 10.1109/CVPR.2018.00474.

[11] P. Warden and D. Situnayake, "TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers," 1st ed. Sebastopol, CA, USA: O'Reilly Media, Dec. 2019, ISBN: 978-1492052043.

[12] TensorFlow Team, "TensorFlow Lite Guide: Deploy machine learning models on mobile and edge devices," Google LLC, Mountain View, CA, USA, ver. 2.6, 2021. [Online]. Available: https://www.tensorflow.org/lite/guide. [Accessed: Aug. 31, 2025].

[13] M. A. Al-Garadi, A. Mohamed, A. K. Al-Ali, X. Du, I. Ali, and M. Guizani, "A survey of machine and deep learning methods for internet of things (IoT) security," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1646-1685, 3rd Qtr. 2020, doi: 10.1109/COMST.2020.2988293.

[14] A. Kumar, S. Goyal, and M. Varma, "Resource-efficient machine learning in 2 KB RAM for the internet of things," in *Proc. 34th International Conference on Machine Learning (ICML)*, Sydney, Australia, Aug. 6-11, 2017, pp. 1935-1944.

[15] D. Gehrig, A. Rebecq, G. Gallego, and D. Scaramuzza, "EKLT: Asynchronous photometric feature tracking using events and frames," *International Journal of Computer Vision*, vol. 128, no. 3, pp. 601-618, Mar. 2020, doi: 10.1007/s11263-019-01209-w.

[16] R. Manduchi and J. M. Coughlan, "The last meter: blind visual guidance to a target," in *Proc. SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*, Toronto, Canada, Apr. 26 - May 1, 2014, pp. 3113-3122, doi: 10.1145/2556288.2557328.

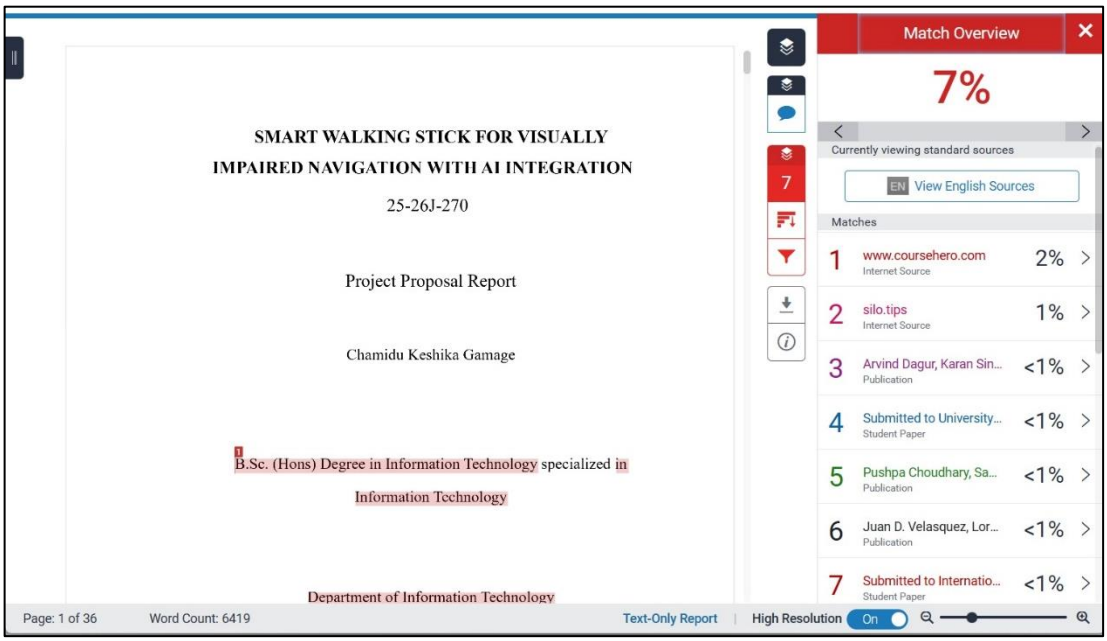
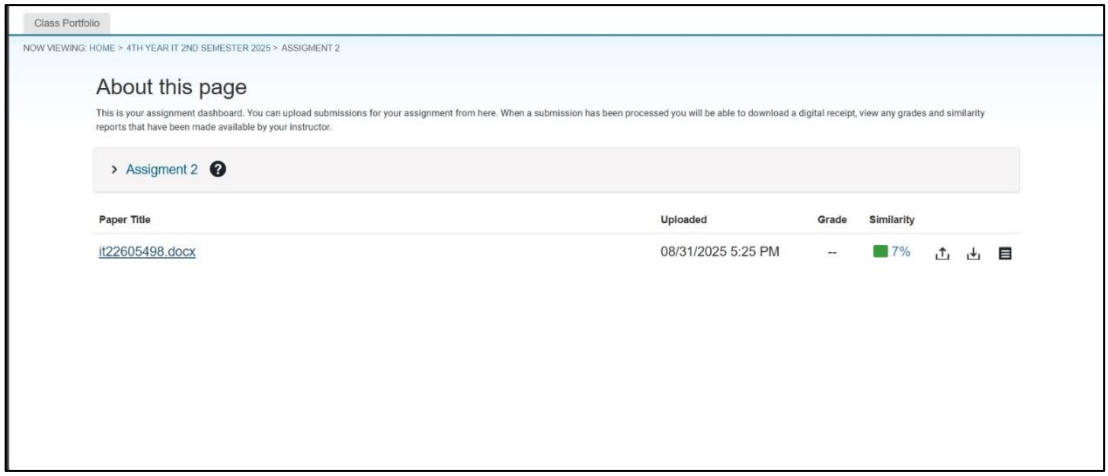
[17] L. A. Jones and S. J. Lederman, "Human Hand Function," New York, NY, USA: Oxford University Press, 2006, ISBN: 978-0195173154.

[18] H. Z. Tan and A. Pentland, "Tactual displays for wearable computing," *Personal and Ubiquitous Computing*, vol. 1, no. 4, pp. 225-230, Dec. 1997, doi: 10.1007/BF01682024.

APPENDICES

Survey Link - https://docs.google.com/forms/d/e/1FAIpQLSd5cBcEZoJJkBGScwZehb8XtkEo5FPxwk3ICwPocjVrx5wuWQ/viewform?usp=header

Plagiarism report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